

Assignment 1 --CDC Diabetes Health Indicator

EDA & CLASSIFICATION REPORT

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Dataset shape: 253680 rows × 22 columns (**source:** CDC BRFSS 2015, UCI ML Repository id=891). KNN/Naive Bayes trained on a **stratified 15,000-row sample**

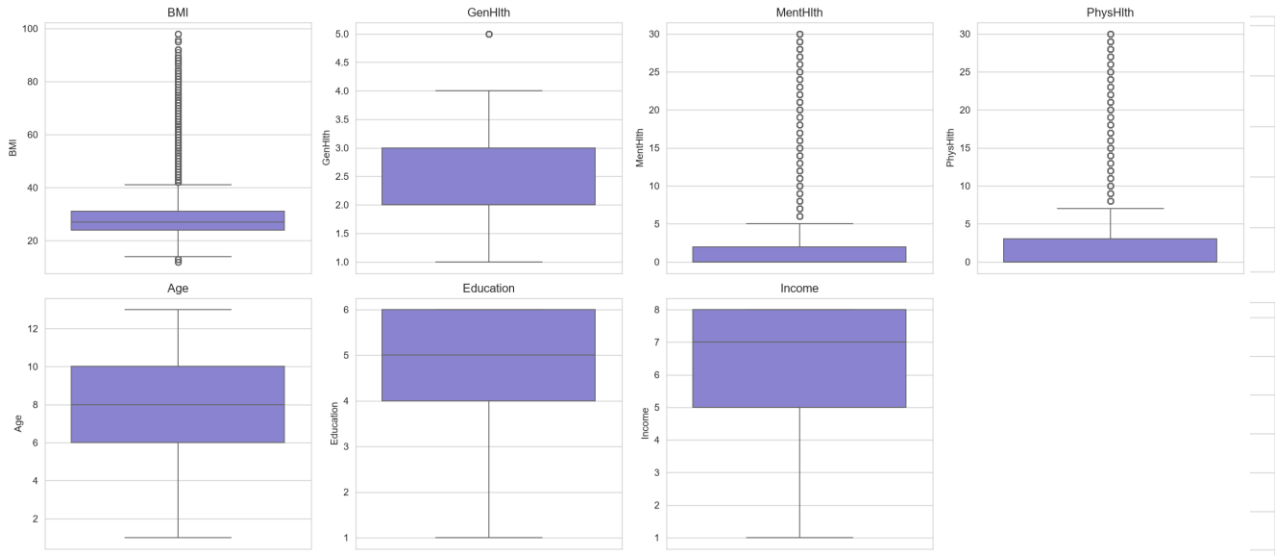
1. Data Quality Report

Missing values were imputed using class-conditional medians before this report was generated.

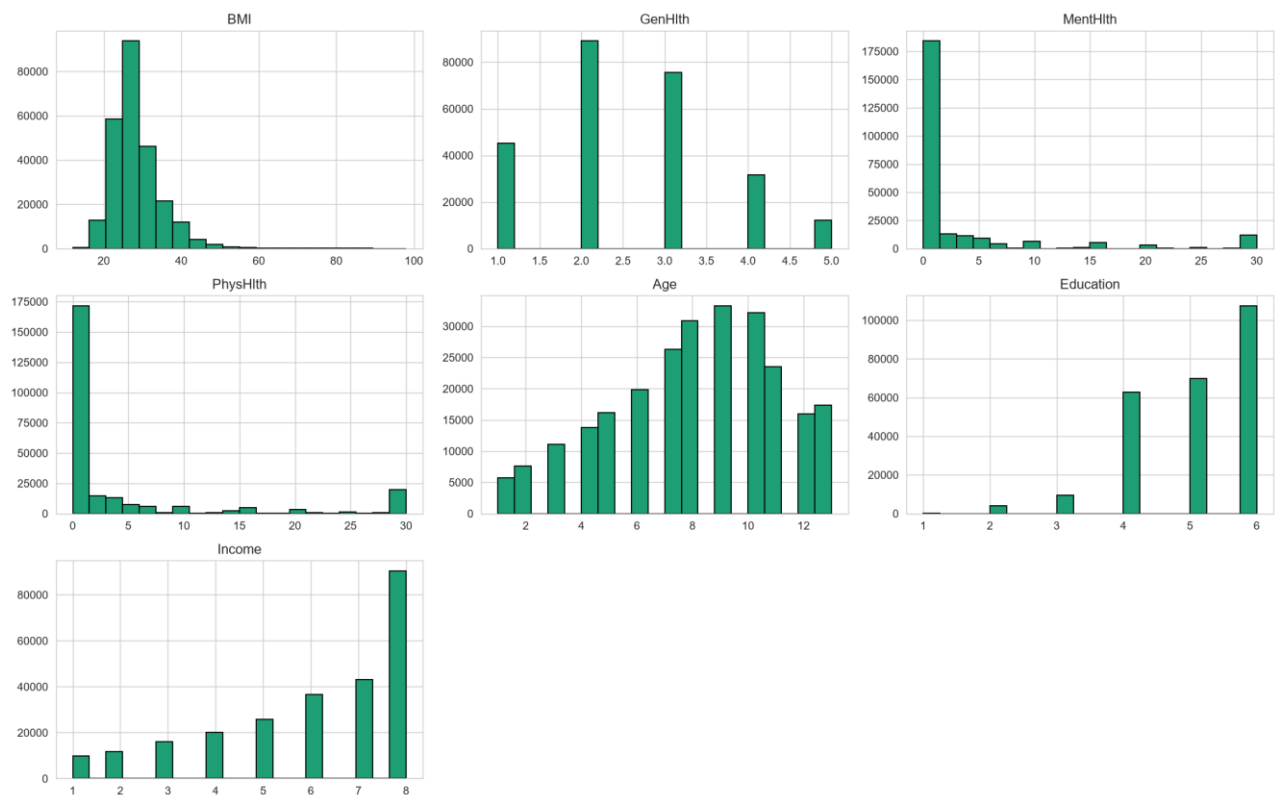
	dtype	cardinality
HighBP	int64	2
HighChol	int64	2
CholCheck	int64	2
BMI	int64	84
Smoker	int64	2
Stroke	int64	2
HeartDiseaseorAttack	int64	2
PhysActivity	int64	2
Fruits	int64	2
Veggies	int64	2
HvyAlcoholConsump	int64	2
AnyHealthcare	int64	2
NoDocbcCost	int64	2
GenHlth	int64	5
MentHlth	int64	31
PhysHlth	int64	31
DiffWalk	int64	2
Sex	int64	2
Age	int64	13
Education	int64	6
Income	int64	8
Outcome	int64	2

2. Visualizations

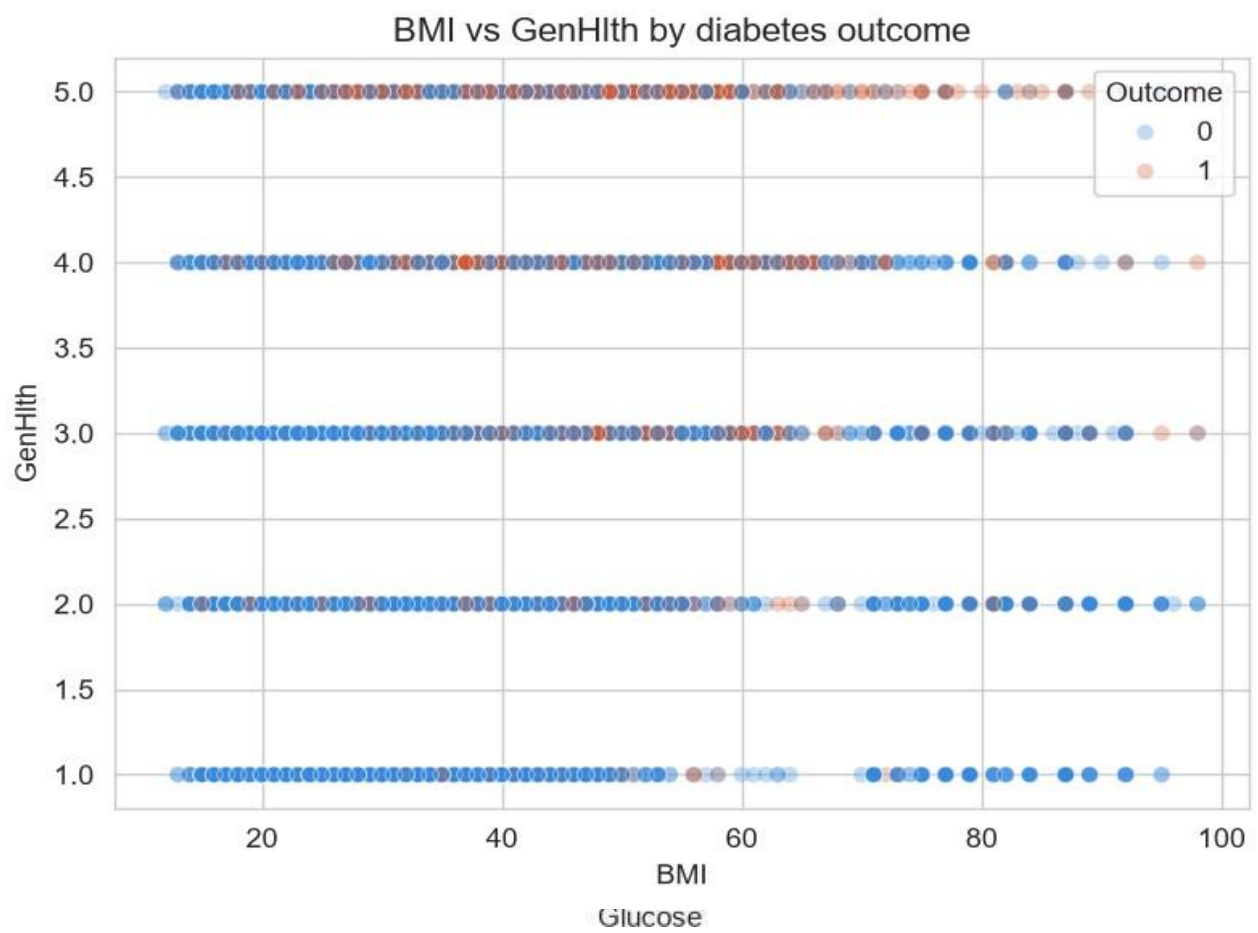
Boxplots



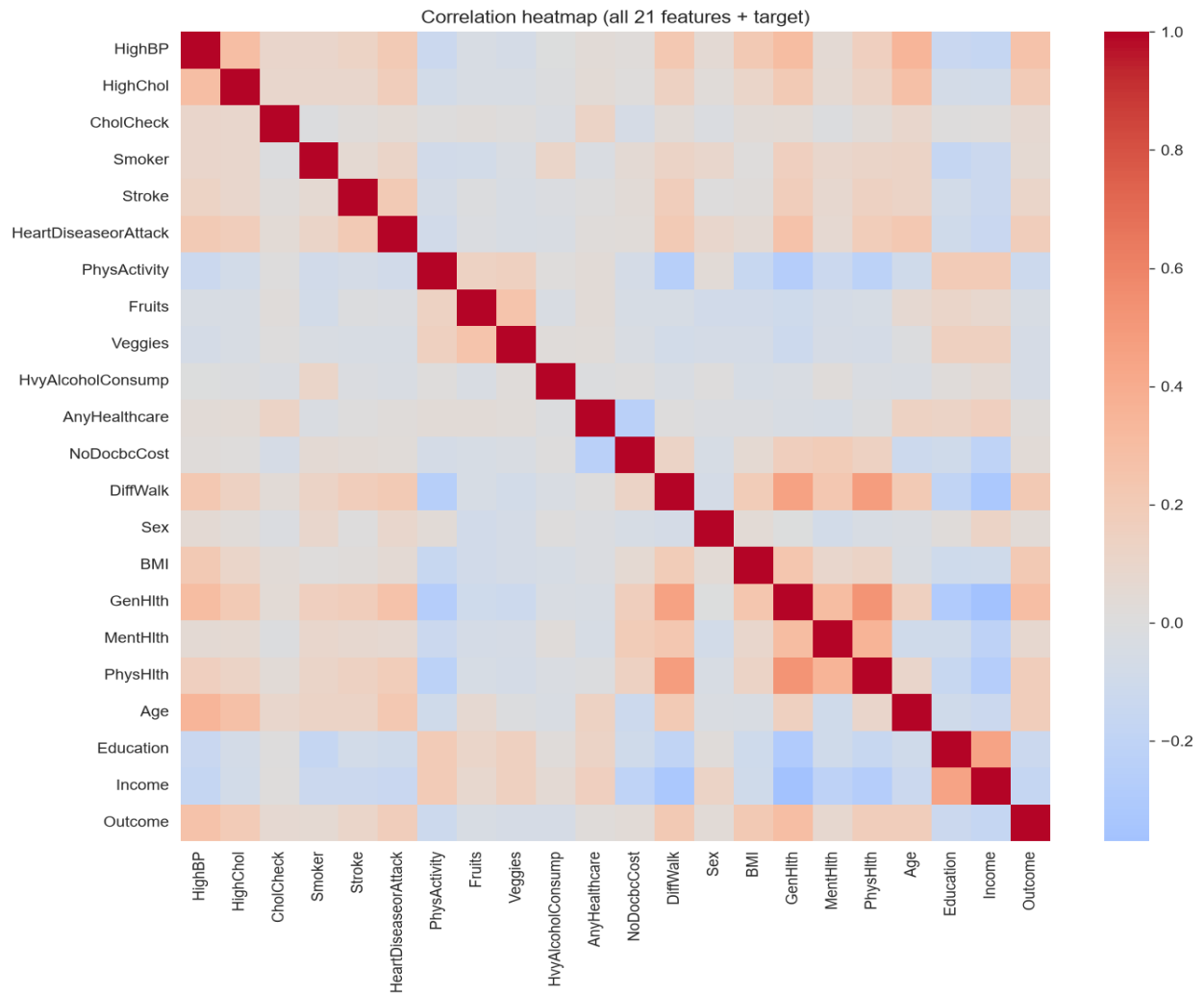
Histograms



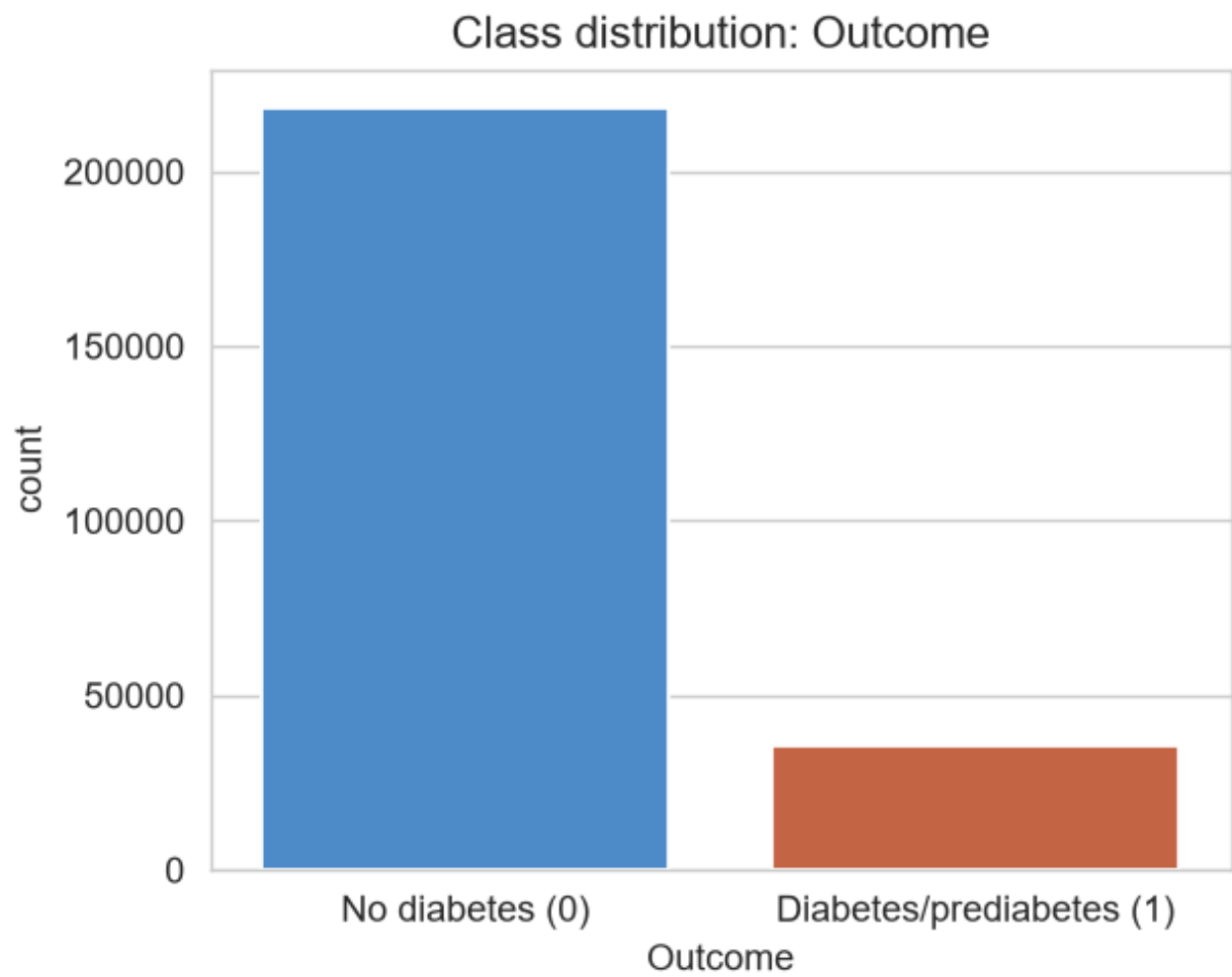
Scatter: Glucose vs BMI



Correlation heatmap



Class Distribution

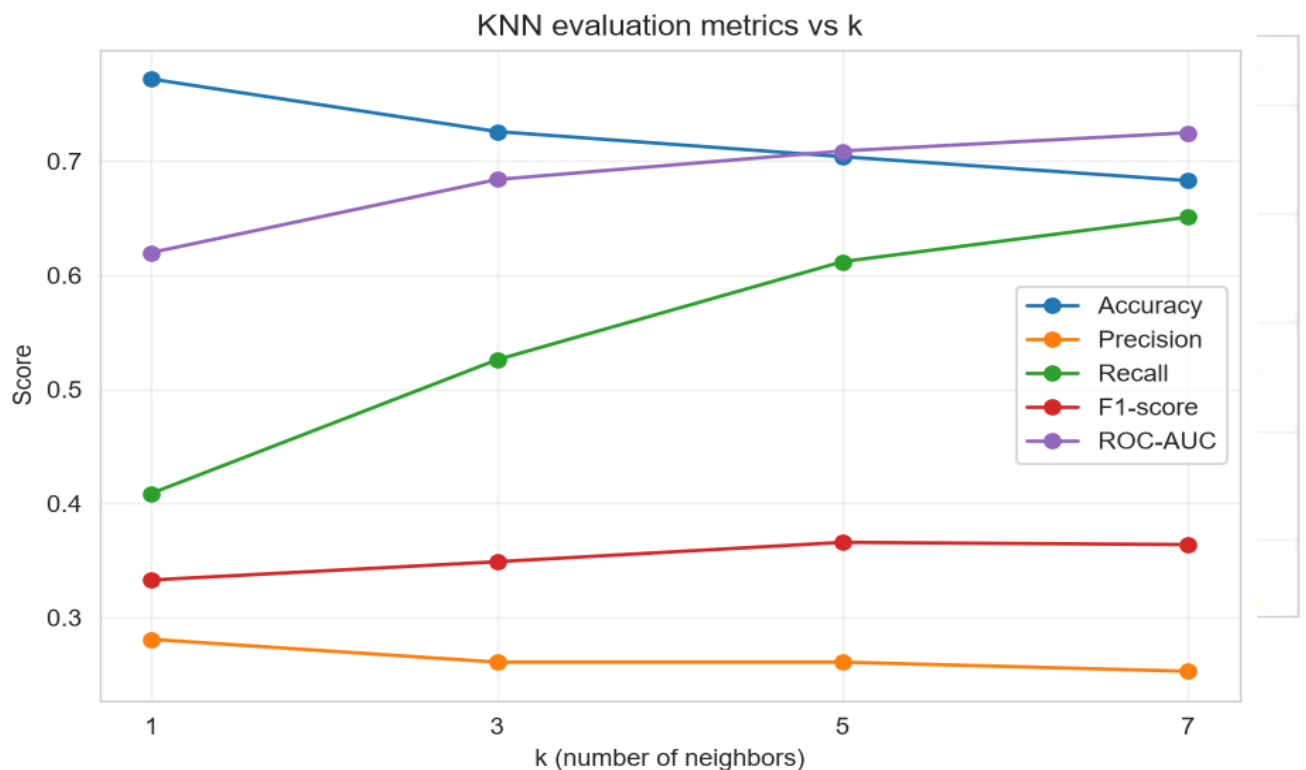


3. Model Results (*KNN vs Naive Bayes*)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN	0.704	0.261	0.612	0.366	0.709
Naive Bayes	0.712	0.29	0.734	0.416	0.781

4. KNN performance across k values

k	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
1	KNN (k=1)	0.772	0.281	0.409	0.333	0.62
3	KNN (k=3)	0.726	0.261	0.526	0.349	0.684
5	KNN (k=5)	0.704	0.261	0.612	0.366	0.709
7	KNN (k=7)	0.683	0.253	0.651	0.364	0.725



5. Best Model Comparison

Best-tuned KNN vs Naive Bayes

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN (k=5)	0.704	0.261	0.612	0.366	0.709
Naive Bayes	0.712	0.29	0.734	0.416	0.781

